BANKING ALGORITHM - SIVARATHINAM S (241401102)

Activities Terminal Mar 20 21:08

sivarathinam@localhost: — /usr/bin/vim banekersigo.c

```
lude<stdio.h>

main() {
int n, r, i, j, k;
printf("enter no of processes:");
scanf("%d", &n);
printf("enter no of resource types:");
scanf("%d", &r);

int alloc[n][r], max[n][r], need[n][r], avail[r];
int finish[n], safe_sequence[n];
int count = 0, safeFound;

printf("enter allocation matrix:\n");
for (i = 0; i < n; i++) {
    for (j = 0; j < r; j++) {
        scanf("%d", &alloc[i][j]);
    }
}

printf("enter max matrix:\n");
for (i = 0; i < n; i++) {
    for (j = 0; j < r; j++) {
        scanf("%d", &max[i][j]);
    }
}

printf("enter available resources:\n");
for (i = 0; i < r; i++) {
    scanf("%d", &avail[i]);
}

for (i = 0; i < n; i++) {
                                                                    1,1          To
```

```
Activities   Terminal                    Mar 20  21:09
sivarathinam@localhost:— /usr/bin/vim bakersalgo.c

        for (i = 0; i < n; i++) {
            for (j = 0; j < r; j++) {
                need[i][j] = max[i][j] -alloc[i][j];
            }
        }

        while (count < n) {
            safeFound = 0;
            for (i = 0; j < n; i++) {
                if (finish[i] == 0) {
                    int safe = 1;
                    for (j = 0; j < r; j++) {
                        if (need[i][j] > avail[j] {
                            safe = 0;
                            break;
                        }
                    }
                }
            }
        }

        if (!safeFound) {
            printf("system is unsafe state! may occur the deadlock \n");
        }
    }
    if (safeFound || count == n) {
        printf("safe sequence is:";
        for (i = 0; i = n; i++)
            printf("%d ", safe_sequence[i]);
    }
}
return 0;
                                                                    81,1          Bot
```

Output :

```
Activities   Terminal                    Mar 20  21:12
sivarathinam@localhost:~

3  sivaathinam@localhost:~
4  history
4  histoggo
ivaathinam@lhost:~$ bankersalgo.c
varathinam@lhost:~$ bankersalgo.c
varathinam@lhost:~$ bankersalgo.c
varathinam@lhost:~$ gcc bankersalgo.-o bankers
nter no of processes:3
nter no of resource types:3
nter allocation matrix:
  4 0 2
  5 2 0
  1 1 0
  2 1 0
nter max matrix:
  5 0 2
  1 0 2
  2 2 2
  9 1 2
ter max matrix:
  5 0 2
  3 2 2
  9 1 2



nter available resources:
  8 8



fe sequencce is:P0 P1 P[caleb@localhost ~]$
varathinam@localhost:~$
```